

AgroShield AI - Technical Summary

Project Technology Stack Overview & Deployment Architecture

1. Core AI & Backend Architecture

- **Deep Learning Framework:** TensorFlow & Keras for neural net modeling.
- **Image Classification Model:** MobileNetV2 (Transfer Learning) with frozen ImageNet weights (**89.47% accuracy**).
- **Image Preprocessing:** NumPy & Pillow (PIL) for scaling, cropping, and RGB normalization.
- **Web Server API:** Python Flask for RESTful endpoint routing.
- **Data Storage:** Thread-Safe JSON Document Store for fast, lightweight local database persistence.

2. Interactive Frontend & User Interface

- **Design Language:** Responsive Glassmorphic Dark UI using HTML5 & Vanilla CSS3.
- **Geospatial Mapping:** Leaflet.js for interactive surveillance heatmaps and custom status markers.
- **Data Visualization:** Chart.js for outbreak distribution, timeline trends, and 7-day weather projections.
- **Voice Assistant (USP):** Web Speech API (*speechSynthesis*) for multi-language spoken advisories (English, Hindi, Spanish, Swahili).
- **Performance Optimization:** Custom Fetch Abort Controller enabling instant 2.5s client-side fallback mode.

3. Deployment & Cloud Infrastructure

- **Frontend Hosting:** Vercel for high-speed, static asset delivery.
- **Backend Hosting:** Render / Railway for containerized Python/Flask web services.
- **Routing Configuration:** Custom *vercel.json* rewrites for seamless front-to-back proxying.
- **Version Control:** Git & GitHub for code backups and automated Vercel CI/CD pipelines.